AP CHEMISTRY · THE UNIVERSAL STUDY GUIDE

Targeting a 3, 4, or 5 — every concept tagged for your tier

Tier badges: [3] foundational · [4] adds for 4+ · [5] adds for 5

Targeting a 4 → study [3] AND [4] items. Targeting a 5 → study all three tiers.

PAGE 1 | MATH OF 3/4/5 · 9 UNITS · PARTICULATE-LEVEL THINKING

About this exam. AP Chem demands **particulate-level reasoning**. Many MCQs and FRQs ask you to draw or interpret particle diagrams (atoms, ions, molecules) and EXPLAIN macro behavior in terms of micro structure. "Why is X soluble?" → particle-level intermolecular forces answer.

THE MATH OF YOUR TARGET

Target	MCQ (60q, 50%)	FRQ (7q, 50%)	Approx %	Tier
3 (PASS)	**~30 / 60** (~50%)	**~22/46 raw**	~52-56%	[3]
4	**~38 / 60** (~63%)	**~28/46 raw**	~62-67%	[3]+[4]
5	**~46 / 60** (~77%)	**~36/46 raw**	~75-80%	All

Time: 90 min for 60 MCQ (~1:30 each), 105 min for 7 FRQs — **3 long (10 pts each) + 4 short (4 pts each)**. Calculator allowed for FRQ section, NOT MCQ. Reference table provided.

THE 9 UNITS — by exam weight

#	Unit	Weight	Focus
1	Atomic Structure & Properties	7–9%	Moles, isotopes, mass spec, photoelectron spectroscopy, periodic trends
2	Compound Structure & Properties	7–9%	Ionic vs molecular, Lewis structures, VSEPR, hybridization
3	Properties of Substances + Mixtures	10–12%	**IMF (LDF, dipole, H-bond)**, solutions, gas laws
4	Chemical Reactions	7–9%	Types: precipitation, acid-base, redox; net ionic equations
5	Kinetics	7–9%	Rate laws, mechanisms, Arrhenius, catalysts
6	Thermodynamics	7–9%	Enthalpy, calorimetry, Hess's law, bond energies
7	Equilibrium	10–15%	**ICE tables**, K, Q, Le Chatelier, solubility (Ksp)
8	Acids & Bases	11–15%	**pH, pKa, Ka/Kb, buffers, titrations** — heaviest unit
9	Applications of Thermodynamics	7–9%	**Gibbs free energy**, electrochemistry, voltaic + electrolytic cells

KEY INSIGHT **[5] Units 7-9 = ~30% of the exam combined.** Equilibrium + acid-base + thermodynamics applications are interrelated. Master Le Chatelier reasoning + ICE tables + Gibbs free energy connections — they appear together on long FRQs.

PAGE 2 | UNIT 1: ATOMIC STRUCTURE · MASS SPEC · PES

Moles + Avogadro [3]

```
1 mol = 6.022 × 10²³ particles    (Avogadro's number)

moles = mass / molar mass
particles = moles × 6.022 × 10²³

Molar mass found by adding up atomic masses from periodic table.

Common conversions:
  grams → moles → particles (atoms or molecules)
  grams → moles → liters (gas at STP: 22.4 L)
  grams → moles → moles other species (via stoichiometry coefficients)
```

Mass Spectrometry [3]/[4]

- **[3] Reads MASS / charge of ions.** X-axis = mass, y-axis = relative abundance.
- **[3] Multiple peaks = ISOTOPES.** The TALLER peak = more abundant isotope.
- **[4] Calculate average atomic mass**: ($mass_1 \times abundance_1$) + ($mass_2 \times abundance_2$)...

Photoelectron Spectroscopy (PES) [3]/[4]/[5]

```
PES measures BINDING ENERGIES of electrons in atoms/ions.
X-axis: binding energy (typically MJ/mol)
Y-axis: relative # of electrons

PEAK INTERPRETATION:
  Peak HEIGHT = ratio of electrons in that subshell
    (1s = 2 electrons; 2p = up to 6; etc.)
  Peak POSITION (binding energy):
    Higher BE = electrons CLOSER to nucleus (1s > 2s > 2p > 3s ...)

EXAMPLE - SODIUM (Na, 11 electrons):  1s² 2s² 2p⁶ 3s¹
  PES peaks: 1s (highest BE, height 2),
             2s (medium-high, height 2), 2p (lower, height 6),
             3s (lowest, height 1)
```

Periodic Trends [3]/[4]

Trend	Pattern + WHY
Atomic radius [3]/[4]	INCREASES down a group (more shells), DECREASES across a period (more protons pulling electrons in same shell — effective nuclear charge ↑).
Ionization energy [3]/[4]	OPPOSITE of radius. Increases across, decreases down. Why: closer electrons + higher Zeff = harder to remove.
Electronegativity [3]/[4]	Same as IE. Fluorine = highest. Cesium = lowest (excluding noble gases).
Successive IEs [4]/[5]	Big jump between valence shell and core electrons signals number of valence electrons. Mg's IE2 → IE3 jump = 2 valence electrons.

KEY INSIGHT **[5] Effective nuclear charge (Zeff) explains everything.** Zeff = protons attracting an electron, minus shielding from inner-shell electrons. Higher Zeff = electrons held tighter = smaller atom + higher IE + higher EN. Memorize this single concept.

PAGE 3 | UNITS 2-3: BONDING · LEWIS · VSEPR · IMF · SOLUTIONS

Lewis Structures [3]/[4]

```
STEPS:
 1. Count total VALENCE electrons (group # for main-group; +/- for charge)
 2. Identify central atom (least electronegative, except H)
 3. Connect outer atoms to central with single bonds
 4. Distribute remaining electrons as LONE PAIRS to satisfy octets (start outer)
 5. If central atom short on octet → add double/triple bonds
 6. Verify charges (formal charge = valence e⁻ − lone pair e⁻ − ½(bond e⁻))

EXCEPTIONS:
  Less than octet: H (2), B (6), Be (4)
  More than octet: P, S, Cl, etc. (period 3+) — expanded octet
```

VSEPR — geometry from electron domains [3]/[4]

Domains	Bonded	Lone pairs	Geometry	Angle
2	2	0	Linear	180°
3	3	0	Trigonal planar	120°
3	2	1	Bent	<120°
4	4	0	Tetrahedral	109.5°
4	3	1	Trigonal pyramidal	<109.5°
4	2	2	Bent	<109.5°
5	5	0	Trigonal bipyramidal	90°/120°
6	6	0	Octahedral	90°

[4] Hybridization corresponds to # of domains: 2 = sp, 3 = sp^2, 4 = sp^3, 5 = sp^3d, 6 = sp^3d^2.

Polarity — molecular vs bond [3]/[4]

- **Polar BOND [3]** = different electronegativities. $\Delta EN > 0.5$ typically polar.
- **Polar MOLECULE [3]/[4]** = polar bonds + asymmetric geometry. CO_2 has polar bonds but is LINEAR (symmetric) → NONPOLAR.
- **[4] Symmetry test**: if all outer atoms identical AND geometry is symmetric (linear, trigonal planar, tetrahedral, etc. with all same substituents), molecule is NONPOLAR.

Intermolecular Forces (IMF) — strength order [3]/[4]/[5]

```
STRONGEST → WEAKEST:

ION-DIPOLE      ions in polar solvent (e.g., NaCl in water) — strongest
HYDROGEN BOND   H bonded to N, O, or F.  HF, H₂O, NH₃
DIPOLE-DIPOLE   polar molecules attract each other
LONDON DISP.    momentary dipoles in all molecules; only force in nonpolar
                (LDF strength ↑ with electrons / molar mass / surface area)

ALL MOLECULES have LDF.  Polar adds dipole-dipole.  N/O/F-H adds H-bonding.
```

IMF Effects on properties [3]/[4]

- **Higher IMF [3]/[4]** → higher boiling/melting point, higher viscosity, lower vapor pressure.
- **Solubility "like dissolves like" [3]/[4]**: polar in polar, nonpolar in nonpolar.
- **[5] Particle diagrams** for solubility: must show dipoles, ion-dipole interactions, or hydrogen bonds. Not just "dots".

Gas Laws [3]/[4]

```
IDEAL GAS:    PV = nRT     R = 0.0821 L·atm/(mol·K)

COMBINED:    P₁V₁/T₁ = P₂V₂/T₂

PARTIAL PRESSURE (Dalton):  P_total = P₁ + P₂ + ... = X_i × P_total
                            (X_i = mole fraction)

REAL GASES deviate from ideal at:
  HIGH pressure (molecules close, repulsion + attraction matter)
  LOW temperature (slow molecules → IMF significant)
```

PAGE 4 | UNITS 4-5: REACTIONS · STOICHIOMETRY · KINETICS

Reaction Types [3]/[4]

Type	Pattern + spotting
Synthesis [3]	A + B → AB. Two reactants combining.
Decomposition [3]	AB → A + B. One reactant splits.
Single replacement [3]/[4]	A + BC → AC + B. Activity series — more reactive metal displaces less reactive.
Double replacement (precipitation) [3]/[4]	AB + CD → AD + CB. Solubility rules: insoluble product = precipitate.
Acid-base (neutralization) [3]	Acid + base → salt + water (often). Proton transfer.
Redox [3]/[4]	Electrons transfer. Oxidation states change. LEO says GER (Lose e^- = Oxidation, Gain e^- = Reduction).

Net Ionic Equations [4]

```
Process:
 1. Write the molecular equation (with formulas of all compounds)
 2. Break SOLUBLE STRONG ELECTROLYTES into ions (aq form)
 3. Cancel SPECTATOR ions (appear unchanged on both sides)
 4. Result: NET IONIC EQUATION

Examples:
  AgNO₃(aq) + NaCl(aq) → AgCl(s) + NaNO₃(aq)
  Net ionic:  Ag⁺(aq) + Cl⁻(aq) → AgCl(s)
```

Stoichiometry [3]/[4]

```
STOICHIOMETRY ROADMAP:

  grams A   →   moles A   →   moles B   →   grams B
           /MM_A          ×coeff B/coeff A           ×MM_B

LIMITING REAGENT:
  Calculate moles of product from EACH reactant
  The reactant producing LESS = LIMITING (the other is excess)

PERCENT YIELD = (actual / theoretical) × 100%
```

Kinetics — rate laws [3]/[4]

```
Rate = k[A]^m [B]^n      (m, n = orders, NOT coefficients)
Overall order = m + n

DETERMINING ORDERS from initial-rate data:
  Compare 2 trials where ONE concentration varies.
  If [A] doubles + rate doubles → 1st order in A
  If [A] doubles + rate quadruples → 2nd order in A
  If [A] doubles + rate unchanged → 0th order in A

HALF-LIFE for 1st-order:  t₁/₂ = 0.693 / k    (constant!)
Integrated rate laws:
  0th order:  [A] = [A]₀ - kt    (linear plot of [A] vs t)
  1st order:  ln[A] = ln[A]₀ - kt  (linear plot of ln[A] vs t)
  2nd order:  1/[A] = 1/[A]₀ + kt  (linear plot of 1/[A] vs t)
```

Mechanisms [4]/[5]

- **Rate-determining step (RDS) [4]**: SLOWEST step. Rate law from RDS using its reactants.
- **[5] Intermediate** = produced in early step, consumed later. Doesn't appear in overall equation.
- **[5] Catalyst** = lowers activation energy, NOT consumed. Provides alternative pathway.
- **Arrhenius [4]/[5]**: $k = A \cdot e^{(-Ea/RT)}$. Higher T or lower Ea → larger k → faster.

KEY INSIGHT **[4] Connect kinetics to particle behavior.** Higher T → more particles with energy ≥ Ea → more effective collisions → faster rate. Catalyst → alternative pathway with lower Ea → more particles meet new threshold → faster.

PAGE 5 | UNIT 6: THERMOCHEMISTRY — ENTHALPY · CALORIMETRY

Enthalpy Basics [3]/[4]

```
ΔH = enthalpy change at constant pressure

EXOTHERMIC:  ΔH < 0  (releases heat to surroundings)
ENDOTHERMIC: ΔH > 0  (absorbs heat from surroundings)

ΔH_rxn = (sum bond energies BROKEN) − (sum bond energies FORMED)
     = [reactants' bonds energy] − [products' bonds energy]

ΔH_rxn = Σ ΔHf(products) − Σ ΔHf(reactants)        (Hess's law)
  where ΔHf = standard enthalpy of formation
  (ΔHf of element in standard state = 0)
```

Calorimetry [3]/[4]

```
q = m · c · ΔT
  q  = heat (J)
  m  = mass (g)
  c  = specific heat capacity (J/g°C)
  ΔT = T_final − T_initial (always final minus initial!)

COFFEE-CUP CALORIMETRY (constant P, dilute solutions):
  q_rxn = −q_solution = −m_soln · c_soln · ΔT
  For aqueous: c ≈ 4.18 J/g°C, density ≈ 1 g/mL

BOMB CALORIMETRY (constant V):
  q_v = q_rxn = −C_cal · ΔT     (C_cal = heat capacity of bomb)
```

Hess's Law [3]/[4]/[5]

[3]/[4] If reactions sum to give the target reaction, their ΔH values sum too. When manipulating equations:

- Reverse a reaction → flip the sign of ΔH.
- Multiply a reaction by n → multiply ΔH by n.
- Add reactions → add ΔH values.

Heating Curves + Phase Changes [3]/[4]

- **[3] Flat segments** = phase change (heat goes into breaking IMF, not raising T).
- **[3] Sloped segments** = T changing within phase ($q = mc\Delta T$).
- **[4] Total q for heating + melting + heating + boiling + heating** = sum of each segment.

Bond Energies [4]/[5]

```
ΔH_rxn ≈ Σ(bond energies of bonds BROKEN) − Σ(bond energies of bonds FORMED)

REMEMBER: breaking bonds COSTS energy (positive); forming bonds RELEASES (negative).

Trends in single bond energies:
  H-F > H-Cl > H-Br > H-I    (smaller atoms → stronger bonds)
  Triple > Double > Single   (more shared electrons → stronger)
```

PAGE 6 | UNIT 7: EQUILIBRIUM — K · Q · ICE · LE CHATELIER

Equilibrium constant K [3]/[4]

```
For aA + bB ⇌ cC + dD:
  K_c = [C]^c [D]^d / ([A]^a [B]^b)
K_p uses partial pressures (gases):  K_p = K_c × (RT)^Δn

K >> 1:  products favored (lies right)
K ≈ 1:  significant amounts of both
K << 1:  reactants favored (lies left)

Pure solids + liquids are NOT included in K expressions.
```

Q (Reaction Quotient) [3]/[4]

```
Same form as K but uses CURRENT concentrations (not equilibrium).

Q < K  →  too much reactant. Reaction shifts RIGHT (forward).
Q = K  →  at equilibrium.
Q > K  →  too much product. Reaction shifts LEFT (reverse).
```

ICE Tables [3]/[4]/[5]

```
EXAMPLE:  N₂(g) + 3H₂(g) ⇌ 2NH₃(g),  K_c = 0.50
Initial: [N₂] = 1.0, [H₂] = 1.0, [NH₃] = 0

               N₂      H₂      NH₃
Initial:       1.0     1.0     0
Change:        -x      -3x     +2x
Equilibrium:   1.0-x   1.0-3x  2x

Plug into K:  (2x)² / [(1.0-x)(1.0-3x)³] = 0.50
Solve for x → equilibrium concentrations.

SHORTCUT [4]: if K is very small (<10⁻³), assume x is tiny → 1.0-x ≈ 1.0
```

Le Chatelier's Principle [3]/[4]

Stress	System response
Add reactant [3]	Shifts RIGHT (away from added).
Add product [3]	Shifts LEFT.
Increase pressure (decrease volume) [3]/[4]	Shifts to FEWER moles of gas.
Decrease pressure (increase volume) [3]/[4]	Shifts to MORE moles of gas.
Increase T (exothermic rxn) [4]	Shifts LEFT (away from heat). For endothermic: shifts RIGHT.
Add catalyst [3]/[4]	**No shift!** Speeds up forward + reverse equally.
Add inert gas at constant V [5]	**No shift** — doesn't change partial pressures of reacting gases.

Solubility (Ksp) [4]/[5]

```
For salt MX(s) ⇌ M⁺(aq) + X⁻(aq):
  Ksp = [M⁺][X⁻]    (solid not included)

For Ag₂CrO₄(s) ⇌ 2Ag⁺(aq) + CrO₄²⁻(aq):
  Ksp = [Ag⁺]² [CrO₄²⁻]
  If solubility = s, then [Ag⁺] = 2s, [CrO₄²⁻] = s → Ksp = 4s³

COMMON ION EFFECT [5]: adding ion already in solution → DECREASES solubility
```

KEY INSIGHT **[5] Le Chatelier ↔ Q vs K connection.** Stress changes Q. $Q < K$ → forward shift (Q increases toward K). $Q > K$ → reverse shift. This is the rigorous way to predict shifts beyond memorized rules.

PAGE 7 | UNIT 8: ACIDS · BASES · BUFFERS · TITRATIONS

pH + pOH essentials [3]/[4]

```
pH = −log[H⁺]                pOH = −log[OH⁻]
pH + pOH = 14    (at 25°C)
Kw = [H⁺][OH⁻] = 1.0 × 10⁻¹⁴    (water autoionization)

STRONG ACIDS (fully ionize): HCl, HBr, HI, HNO₃, HClO₄, HClO₃, H₂SO₄
STRONG BASES: Group 1 hydroxides (NaOH, KOH...), Ca/Sr/Ba(OH)₂

WEAK ACIDS:  partial ionization. Use Ka.
  Ka = [H⁺][A⁻] / [HA]        pKa = −log(Ka)
WEAK BASES:  use Kb. Ka × Kb = Kw (conjugate pair)
```

Calculating pH [3]/[4]

```
STRONG ACID:  pH = −log(initial concentration)
  e.g., 0.1 M HCl → pH = −log(0.1) = 1

WEAK ACID HA at concentration C:
  Ka ≈ x² / C    (assuming x << C)
  x = [H⁺] = √(Ka·C)
  pH = −log(x)

WEAK BASE B (similar process with Kb)
```

Buffers [3]/[4]/[5]

Buffer = weak acid + its conjugate base (or weak base + conjugate acid). Resists pH change when small amounts of acid/base added.

```
HENDERSON-HASSELBALCH:
  pH = pKa + log([A⁻]/[HA])

When [A⁻] = [HA]:  pH = pKa  (the buffer's "sweet spot")

BUFFER CAPACITY [5]: maximum when [A⁻] ≈ [HA]. Higher concentrations = greater capacity.

ADDING STRONG ACID to buffer: H⁺ reacts with A⁻ → HA. Use STOICHIOMETRY first,
  then plug new [HA] + [A⁻] into Henderson-Hasselbalch.
```

Titrations [3]/[4]/[5]

Type	Curve features	pH at equivalence
Strong acid + Strong base [3]/[4]	Vertical jump near eq. pt.	pH = 7
Weak acid + Strong base [4]/[5]	Buffer region before eq.; gradual rise.	pH > 7 (basic salt)
Strong acid + Weak base [4]/[5]	Buffer region after eq.; gradual fall.	pH < 7 (acidic salt)

```
HALF-EQUIVALENCE POINT (HE Pt): half of weak acid neutralized.
  [HA] = [A⁻]  →  pH = pKa  (THIS IS WHERE pKa COMES FROM ON A CURVE)

EQUIVALENCE POINT: moles acid = moles base.
  For weak acid + strong base: salt of conjugate base hydrolyzes → pH > 7
  For strong + strong: pH = 7

INDICATORS [4]: pick one whose pKa near eq. pt. pH
  Phenolphthalein (pKa ~9): use for weak acid + strong base
  Methyl orange (pKa ~4): use for strong acid + weak base
```

KEY INSIGHT **[4] On a titration curve, the half-equivalence point is your friend.** It directly reads pKa = pH at HE pt. This is a frequent FRQ part.

PAGE 8 | UNIT 9: GIBBS FREE ENERGY · ELECTROCHEMISTRY

Gibbs Free Energy [3]/[4]/[5]

```
ΔG = ΔH − TΔS

ΔG < 0:  spontaneous (thermodynamically favorable)
ΔG > 0:  non-spontaneous (favorable in REVERSE)
ΔG = 0:  at equilibrium

ΔS = entropy change = (sum products) − (sum reactants)
  More moles of gas → larger ΔS (positive)
  Solid → liquid → gas: ΔS positive
  More complex molecules → higher S

ΔG° = −RT·ln(K)        (links thermodynamics to equilibrium!)
  K > 1 → ΔG° < 0 (products favored)
```

Spontaneity decision matrix [4]/[5]

ΔH	ΔS	Result	Example
−	+	**Spontaneous at all T**	Combustion (exothermic + more gas)
+	−	**Never spontaneous**	Reverse of combustion
−	−	**Spontaneous at LOW T**	Freezing of water
+	+	**Spontaneous at HIGH T**	Melting/boiling, decomposition

Voltaic (Galvanic) Cells [4]/[5]

```
CONVERTS chemical energy → electrical energy (spontaneous redox).

ANODE = oxidation (loses electrons)  — electrons flow OUT
CATHODE = reduction (gains electrons) — electrons flow IN
Mnemonic:  AnOx (Anode = Oxidation), RedCat (Reduction at Cathode)

E°_cell = E°_cathode − E°_anode      (use REDUCTION potentials for both!)
  E°_cell > 0  →  spontaneous

ΔG° = −nFE°_cell        F = 96,485 C/mol

SALT BRIDGE: maintains charge balance. Cations migrate to cathode, anions to anode.
```

Electrolytic Cells [5]

- **[5] OPPOSITE of voltaic.** Electrical energy DRIVES non-spontaneous reaction ($E° < 0$).
- **[5] Used for**: electroplating, refining metals, producing reactive metals (Na from NaCl).
- **[5] Faraday's law**: moles of electrons = current (A) × time (s) / F. Then convert via stoichiometry to grams plated.

KEY INSIGHT **[5] The 3 great connections.** $\Delta G° = -RT \cdot \ln(K)$ [thermodynamics ↔ equilibrium]. $\Delta G° = -nFE°$ [thermodynamics ↔ electrochemistry]. K = e^(nFE°/RT) [equilibrium ↔ electrochemistry]. These are the unit-9 capstone — expect a long FRQ that hops among all three.

PAGE 9 | FRQ STRATEGY · THE 7-QUESTION FORMAT · COMMON TRAPS

FRQ structure [3]/[4]

3 LONG FRQs (10 pts each) + 4 SHORT FRQs (4 pts each) = 46 points total. Calculator + reference table allowed. ~105 minutes.

Type	Topic + recipe
Long Q1	Lab-based. Apparatus, data interpretation, error analysis. Often calorimetry, kinetics, or titration.
Long Q2	Heavy stoichiometry/calculations. Multi-step. Usually involves moles, concentrations, equilibrium.
Long Q3	Particulate-level reasoning. "Draw a particle diagram showing...". Connect macro behavior to micro structure.
Short Q4-7	Each focused on ONE concept. Variety: organic naming, chromatography, redox, periodicity, brief calculation.

Always-do FRQ habits [3]/[4]/[5]

- **[3] Show all work** — graders give partial credit. Naked answer = often 0 even if correct.
- **[3] Include UNITS on every numerical answer.** Missing units = 0.
- **[3] Use sig figs** — usually 2-3 in answer. 4+ rounded down 1 point.
- **[4] Particulate diagrams**: draw clear circles (atoms), use different colors/shading for different elements, show appropriate ratios. NO STICK-FIGURE atoms.
- **[4] Connect math to chemistry**: don't just calculate ΔG; explain what it MEANS for spontaneity in context.
- **[5] Justify with explicit reasoning**: "Because Zeff increases across the period, IE increases, so K is more easily ionized than Ar."

Top point-killers [3]/[4]/[5]

- **[3] Wrong sign on ΔH or ΔS.** Exothermic = NEGATIVE ΔH.
- **[3] Forgetting water/spectator ions in net ionic equations.**
- **[4] Confusing K_c and K_p** for gas reactions. $K_p = K_c \cdot (RT)^{\Delta n}$.
- **[4] Mixing up cathode and anode in electrochemistry.** Anode = oxidation, Cathode = reduction. Use AnOx-RedCat mnemonic.
- **[4] Forgetting to subtract initial T from final T (not the other way).** $\Delta T = T_f - T_i$.
- **[5] Not justifying particulate diagrams.** Drawing alone is half the answer; explanation makes it whole.
- **[5] Plugging into Henderson-Hasselbalch BEFORE neutralization stoichiometry.** When acid/base added to buffer, react FIRST, THEN apply HH.

PAGE 10 | SCORE LADDER · TIER CHECKLISTS · THE ONE THING

THE SCORE LADDER

For a 3 (Pass)	For a 4	For a 5
MCQ: **30-35 / 60**	MCQ: **38-43 / 60**	MCQ: **46-52 / 60**
Stoichiometry, Lewis structures, simple gas laws.	Add: VSEPR + IMF + kinetics + simple equilibrium.	Master ICE tables, buffers, Gibbs ↔ K ↔ E°.
Calculate pH for strong acids/bases.	Calculate pH for weak acid + buffers.	Buffer + titration + half-equivalence pKa identification.
Identify exo/endothermic, simple ΔH calc.	Hess's law + bond energies + calorimetry.	Gibbs free energy + spontaneity matrix + thermo-equilibrium link.
Particle diagrams: simple atom + ions.	Particle diagrams + IMF interactions explained.	Particle diagrams justifying solubility, IMF, polarity.

TIER-SPECIFIC TEST-DAY CHECKLISTS

For a 3 — focus only on these:

- ✓ MCQ: 30+ correct.
- ✓ Stoichiometry: grams ↔ moles ↔ molecules; limiting reagent.
- ✓ Lewis structures + VSEPR for simple molecules (≤4 domains).
- ✓ Identify IMF type: LDF / dipole-dipole / H-bond / ion-dipole.
- ✓ Strong acid pH: pH = −log[H+].
- ✓ $q = mc\Delta T$ calorimetry.
- ✓ Show all work + units + sig figs.

For a 4 — add these on top:

- ✓ MCQ: 38+ correct.
- ✓ ICE tables for equilibrium + simple Le Chatelier predictions.
- ✓ Henderson-Hasselbalch for buffer pH.
- ✓ Half-equivalence point = pKa on titration curves.
- ✓ $\Delta G = \Delta H - T\Delta S$ spontaneity decision matrix.
- ✓ Anode = oxidation, Cathode = reduction (AnOx-RedCat).
- ✓ Particulate diagrams showing IMF interactions.

For a 5 — add these on top:

- ✓ MCQ: 46+ correct.
- ✓ Three connections: $\Delta G^\circ = -RT \cdot \ln(K)$ AND $\Delta G^\circ = -nFE^\circ$ AND K ↔ E°.
- ✓ React acid/base with buffer FIRST, then plug into HH.
- ✓ Common ion effect on solubility.
- ✓ Mechanism analysis: identify intermediates + RDS.
- ✓ Justify particulate diagrams with explicit Zeff or IMF reasoning.
- ✓ Connect macro observations to particulate-level explanations.

THE ONE THING THAT MATTERS MOST [3][4][5]: EXPLAIN MACRO BEHAVIOR WITH PARTICLES. AP Chem rewards students who can connect what they SEE (boiling point, solubility, color change, equilibrium shift) to what's happening at the PARTICLE level (IMF strength, Zeff, electron transfer, collision frequency). "NaCl dissolves in water" earns 0. "NaCl dissolves because polar water molecules orient their dipoles around Na+ and Cl- ions, creating ion-dipole interactions stronger than the ionic bonds in the solid lattice" earns full credit. **Always: particles → macro property.**

If you do nothing else: master ICE tables, the ΔG/K/E° connections, and ALWAYS explain macro with particle reasoning.

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